Yifan Zhu

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Research Profile

Computer Science Ph.D. candidate building **trustworthy LLM agents that reason about what other agents believe, know, and have misunderstood**. My work spans **LLM post-training, reasoning, and alignment** (friction-aware objectives benchmarked against DPO), **agentic multi-agent systems and their evaluation** (benchmarks, calibration, debiased LLM-as-judge), **neuro-symbolic reasoning** that couples LLMs with dynamic epistemic logic and executable meaning representations, and **human–AI collaboration** grounded in carefully annotated multimodal data. I lead projects end to end, from formalization and data construction through multi-GPU training, rigorous evaluation, and open-source prototypes, with first-author work at **EMNLP, SIGDIAL, and LREC**. Seeking a **Summer 2027 research internship** in agentic AI, agent evaluation, or LLM post-training and alignment.

Education

Brandeis University *Expected Spring 2028*
Ph.D. in Computer Science
- **Advisor:** Prof. James Pustejovsky

Georgetown University *2019 – 2021*
M.S. in Computational Linguistics

University College London *2015 – 2016*
M.A. in Linguistics

Xiamen University *2011 – 2015*
B.A. in Philosophy, Chinese Language and Literature

Research Experience

Project: LLM Post-Training for Epistemic Alignment (FPO) *Oct 2025 – Present*
- Designed **friction-aware post-training objectives** that condition language-model interventions on instance-level epistemic risk rather than preference labels alone, and benchmarked them against **DPO**; trained with 4-bit QLoRA on **Qwen3.5-27B** and adapted from synthetic Minecraft interactions to human HCRC MapTask dialogue, FAR and FTR raised MapTask intervention F1 from **0.344 to 0.416–0.417**, cut warranted-context ECE from **0.432 to 0.178–0.227**, and were far more stable across three runs (F1 s.d. **0.023 / 0.009** vs. **0.106** for DPO). (**EMNLP 2026**)
- Built a **rigorous, trustworthy evaluation stack**: **order-debiased LLM-as-judge** scoring, calibration (ECE) and Brier analysis, **bootstrap confidence intervals**, and **multi-seed evaluation**, on an end-to-end multi-GPU **PyTorch / Hugging Face** (TRL, PEFT) pipeline spanning GPT-5 belief-state extraction, supervision generation, QLoRA/DPO training, and vLLM inference.
- Formalized a four-identifier **structured Theory-of-Mind representation** of first- and second-order referential beliefs between dialogue participants, making silent epistemic divergence directly computable; adding the second-order channel raised held-out misunderstanding-detection recall from **26.1% to 65.2%**, with human agreement of $\kappa =$ **0.80/0.77** on inferred second-order beliefs. (**EMNLP 2026**)
- Extended FPO to **perceptually asymmetric dialogue**; across GPT-5, Qwen3.6-35B, and Llama-4-Scout, participant-specific reasoning reached **0.903–0.955 F1** vs. **0.247–0.434** under omniscient access. (**SIGDIAL 2026**)

Selected publications: Mind the Gap: Theory-of-Mind-Grounded Friction for Epistemic Alignment (EMNLP 2026, first author); *From Propositional to Perceptual Asymmetry: Extending FPO to Asymmetric Partial Information Dialogue* (SIGDIAL 2026, first author).

Project: Agentic Multi-Agent LLM Decision-Making under Distributed Evidence *Oct 2025 – Present*
- Showed on **HiddenBench** hidden-profile tasks that **multi-agent LLM groups** (Qwen3.8-27B, Gemma-4-31B, Llama-4-Scout) can converge **unanimously on incorrect decisions** despite complete evidence and citing every private fact; built the **multi-agent simulation and evaluation harness** on vLLM (tensor-parallel) with controlled ablations of prior answers and peer messages.
- Diagnosed agent failure modes: retained initial answers block correction (removing them lifts decoy-choosers'

accuracy **15.6 pts**); failed discussions propagate errors, dropping fully informed readers from **59.7%/90.3% to 0%** accuracy (Qwen/Gemma) after one round.

- Designed a **DEL-inspired evidence-first agent coordination protocol** (verified disclosure, independent judgments before ranking exchange) that beats free discussion by **+40.4 / +63.9 pts** on Qwen (main / additional tasks), beats Exchange-then-Decide by **+17.6 pts** on Gemma, and uses ∼**half the tokens** of free discussion.

Selected publications: Leveraging Shared Evidence: Improving Judgment in LLM Groups (under review, ARR Oct 2026, first author).

Project: Human–AI Belief Tracking & Neuro-Symbolic Common Ground *Mar 2024 – Present*

- Introduced the **Distributed Partial Information Puzzle (DPIP)**, a collaborative **multi-agent benchmark** for evaluating LLM and symbolic belief tracking when participants hold disjoint information; released a multimodal dataset of **40 participants across 10 groups** with aligned annotations of speech, gesture, object interaction, and action (Zenodo).
- Led **data-centric** dataset design, annotation, and adjudication for common-ground and friction corpora, including schema development, annotator training, inter-annotator validation, and quality control.
- Developed an executable **evidence-based dynamic epistemic logic** framework, a **neuro-symbolic** layer that converts multimodal observations and dialogue events into formal updates over each participant's beliefs, intentions, and common ground.
- Implemented the block-description transfer component of **TRACE**, a real-time multimodal common-ground tracking system demonstrated live at **NAACL 2025** (open source), fusing multiparty audio and video evidence for symbolic reasoning.

Selected publications: Distributed Partial Information Puzzles (LREC 2026, first author); *Multimodal Common Ground Annotation for Partial Information Collaborative Problem Solving* (ACL-ISO ISA Workshop 2025, first author); *TRACE: Real-Time Common Ground Tracking* (NAACL 2025 Demo, co-author); *Common Ground Tracking in Multimodal Dialogue* (LREC-COLING 2024, co-author).

Project: Multimodal Perception & Situational Awareness *May 2023 – Present*

- Built live and offline **2D/3D multimodal perception pipelines** (speech, diarization, object detection, pose, gaze, gesture, action) that turn synchronized audio, video, and object-interaction streams into structured events.
- Developed late-fusion situational-awareness models (engagement, joint attention, collaborative behavior) and integrated them into **real-time prototypes and live demos** for **human–AI teaming**.

Selected publications: Modeling Theory of Mind in Multimodal HCI (HCII 2024, first author); *Multimodal Situational Awareness in Collaborative Learning* (HCII 2025, first author); *Speech Is Not Enough: Interpreting Nonverbal Behavior in Multiparty Dialogue* (AAAI 2025, co-author).

Project: Executable Semantics & Structured Meaning Representation (UMR) *2022 – Present*

- Developed **executable semantic representations** that ground **Generative Lexicon event structures** in explicit preconditions and postconditions, supporting **structured symbolic updates** over actions, states, and knowledge.
- Conducted sentence- and document-level Chinese **Universal Meaning Representation** (graph-based semantics) annotation on verb compounds, aspect, and anaphora; proposed solutions **incorporated into UMR community guidelines**.

Selected publications: Lexical Event Models for Multimodal Dialogue (HCII 2024, co-author); *UMR Annotation of Chinese Verb Compounds* (UMR Workshop 2023, co-author).

Technical Skills

- **Post-Training & Alignment:** PyTorch, **Hugging Face** (Transformers, TRL, PEFT), LoRA/QLoRA, DPO, PPO, GRPO, RLHF, reward modeling, domain adaptation
- **Agent Systems & Efficient Inference:** multi-agent simulation and coordination protocols, vLLM (tensor-parallel serving), open-weight deployment, synthetic data, LLM annotation
- **Evaluation & Trustworthy AI:** benchmark design, LLM-as-judge debiasing, calibration/ECE, Brier, bootstrap CIs, multi-seed ablations, IAA
- **Structured Reasoning & Software:** dynamic epistemic logic, executable semantics, UMR; Python, multi-GPU training, Linux, Git; Whisper, pyannote.audio, OpenCV, Detectron2, MMPose

Talks & Professional Service

- **Oral presentation:** *From Propositional to Perceptual Asymmetry: Extending FPO to Asymmetric Partial Information Dialogue*, SIGDIAL 2026.
- **Reviewer:** ACL Rolling Review; LREC 2026; CVPR 2026.